



OPEN Mobility functional status ascertainment in electronic health records using large language models

Xingyi Liu¹✉, Muskan Garg¹, Heling Jia¹, Jennifer St. Sauver², Sandeep R. Pagali³ & Sunghwan Sohn¹

With global aging, assessing functional status is vital for precision medicine. Electronic Health Records (EHRs), particularly unstructured data, hold abundant information on patient mobility. This study explores using Large Language Models (LLMs) to extract and standardize mobility status from unstructured EHR data (i.e., clinical notes). We annotated 600 clinical notes from three healthcare institutions located in southeastern Minnesota and west-central Wisconsin, focusing on expressions of mobility and associated impairment. Leveraging the open-source Llama 3 model, we tested various prompting strategies, including zero-shot, few-shot, and task decomposition, and evaluated their performance. Error analysis showed that while the model sometimes inferred impairments without explicit evidence, most errors were clinically reasonable, often reflecting borderline or ambiguous cases. Our final model achieved a patient-level micro-average F1-score of 0.876 [95% CI 0.858–0.894] for Mobility Extraction and 0.897 [95% CI 0.878–0.917] for Impairment Classification. A secondary analysis counting “clinically reasonable inferences” as correct, performed to assess clinical plausibility, yielded F1-scores of 0.962 [95% CI 0.952–0.971] and 0.948 [95% CI 0.936–0.960], respectively. A local, deterministic setup improved trustworthiness by ensuring consistent outputs, safeguarding privacy, and demonstrating cross-institution generalizability. These findings highlight the feasibility of LLM-based solutions for extracting mobility functional status from unstructured EHR data, supporting both clinical applications and research.

The global demographic landscape is undergoing a shift, with a rapidly increasing proportion of older adults¹. This demographic transition necessitates a fundamental change in healthcare, moving beyond a traditional focus on mortality to prioritize the maintenance of functional status and quality of life^{2,3}. Functional status, broadly defined as an individual's ability to perform activities necessary for daily living, is a powerful predictor of health outcomes, healthcare utilization, and overall well-being, particularly in older populations and those with chronic conditions^{4–6}. Among the various domains of functional status, mobility – the ability to move oneself within an environment – is essential. Mobility limitations are strongly associated with increased risk of falls, hospitalization, institutionalization, reduced social participation, and diminished overall quality of life^{7–9}. Accurate and timely assessment of mobility functional status is therefore essential for effective geriatric care, rehabilitation, and the development of personalized interventions.

Traditionally, mobility functional status has been assessed through standardized questionnaires (e.g., self-reported activity limitations) and performance-based measures (e.g., Timed Up & Go test, gait speed assessment)¹⁰. However, a rich source of information on patient mobility also resides within electronic health records (EHRs)^{11,12}. Clinicians routinely document observations, patient reports, and assessments related to mobility in clinical notes. Unfortunately, this information is typically recorded in unstructured or semi-structured text, making it difficult to extract and analyze systematically. The complexity, variability, and nuanced language used in clinical documentation pose a significant challenge to traditional data extraction techniques. This represents a major barrier to leveraging the wealth of information contained within unstructured EHRs to improve the understanding and management of mobility limitations. Furthermore, manual chart review, the

¹Department of AI and Informatics, Mayo Clinic, Rochester, MN 55905, USA. ²Department of Quantitative Health Sciences, Mayo Clinic, Rochester, MN 55905, USA. ³Department of Medicine, Mayo Clinic, Rochester, MN 55905, USA. ✉email: liu.xingyi@mayo.edu

traditional method for extracting information from unstructured text, is prohibitively time-consuming and expensive for large-scale research or routine clinical use¹³.

This study addresses this challenge by employing Large Language Models (LLMs) to automatically ascertain and standardize mobility functional status from unstructured EHR data, and demonstrates the feasibility of this approach for clinical applications. LLMs, with their advanced capabilities in natural language understanding and generation, offer a powerful means of processing and interpreting complex textual data^{14,15}. Prior research has demonstrated the potential of natural language processing (NLP) techniques, including earlier machine learning approaches and rule-based systems, for extracting information related to functional status and mobility from clinical text^{16,17}. For example, Bales et al. modified an existing NLP system to automate the assignment of selected International Classification of Functioning, Disability, and Health (ICF) codes¹⁸ by updating the lexicon and coding table¹⁹. Agaronnik et al. employed the ClinicalRegex NLP software to search EHRs for functional status documentation, utilizing an ontology with five keyword categories to identify pertinent information²⁰. Newman-Griffis et al. applied NLP techniques to analyze patient functioning information in clinical documents related to federal disability benefit claims from the U.S. Social Security Administration, achieving robust performance with an F1-score exceeding 80% on two datasets²¹. More recently, Fu et al. introduced FedFSA, a hybrid and federated NLP framework designed to extract functional status information from EHRs across multiple healthcare institutions²². These earlier efforts, while showing promise, often faced limitations in terms of generalizability, scalability, and the ability to handle the full complexity and ambiguity of clinical language. Recent advances in LLM architecture and training have demonstrated remarkable performance on a wide range of NLP tasks, including information extraction, question answering, and text summarization, overcoming many of the limitations of prior NLP approaches^{23–25}. These capabilities make LLMs particularly well-suited to the task of extracting structured information from unstructured clinical text, where contextual understanding and the ability to handle ambiguity are crucial. By leveraging the power of an open-source LLM, Llama 3²⁶, we aim to develop a trustworthy, private, and reproducible pipeline that can automatically extract, classify, and standardize mobility-related information from unstructured clinical text.

The contribution of this study is fourfold. First, it develops a comprehensive annotation scheme grounded in the ICF framework to identify and classify expressions related to mobility functional status in clinical notes. Second, it evaluates the performance of Llama 3 in extracting and classifying mobility impairment status by employing various prompt engineering techniques, such as zero-shot learning, few-shot learning, error-informed prompt refinement, and task decomposition. Third, it assesses the generalizability of the LLM-based approach across different healthcare institutions. Finally, it identifies common error patterns and evaluates the overall trustworthiness of the model.

We hypothesize that LLMs with well-constructed prompt engineering strategies can effectively extract mobility functional status information from unstructured EHR data, providing a scalable solution for both clinical practice and research. The successful implementation of this approach has the potential to significantly enhance precision medicine by enabling more timely and personalized interventions, improving the monitoring of patient progress, and facilitating large-scale research on mobility and aging. This, in turn, could lead to improved clinical decision-making, better resource allocation, and ultimately, improved quality of life for older adults and individuals with mobility limitations.

The remainder of this paper is organized as follows. Section 2 presents the results of our experiments, including performance metrics and error analysis. Section 3 discusses the implications of our findings, limitations of the study, and directions for future research. Section 4 details the materials and methods used in this study, including the study population, annotation process, LLM selection, and prompt engineering strategies.

Results

Data analysis

A total of 600 clinical notes were annotated for this study. Table 1 presents the base demographic characteristics of the unique patients associated with these notes. As shown in Table 1, the 600 notes represent 256 unique patients. The overall cohort has a mean age of 75.6 years (SD 13.2). The cohort is sex-balanced, consisting of 137 (53.5%) female patients.

Figure 1a,b present UpSet plots (see Supplementary Fig. S1 for corresponding Venn diagrams), which visualize the frequencies and intersections of the five mobility classes at the section- and note-levels, respectively. In this visualization, the matrix at the bottom defines specific combinations of classes: a filled black dot indicates the presence of a class, and a solid vertical line connecting multiple dots signifies the intersection of those classes. The vertical bar above each column represents the frequency of that specific combination. For example, a vertical line connecting dots for “Changing and maintaining body position” and “Walking and moving” indicates the count of sections containing both classes. At the section-level, the left-hand bar chart shows the total prevalence of each class, with “Changing and maintaining body position” being the most frequent ($n = 853$ sections). The upper bar chart quantifies the intersections; the most common intersection of two or more classes was “Changing and maintaining body position” and “Walking and moving”, occurring in 255 sections. The plot also shows that 44 sections mention “Carrying, moving and handling objects” in isolation (indicated by a single black dot with no connecting lines). A total of 10 sections contained mentions of all five classes, while 2,170 sections mentioned none. At the note-level, the most prevalent class was “Walking and moving” ($n = 222$ notes). The most frequent intersection consisted of all four classes, excluding “Carrying, moving and handling objects”, occurring in 23 notes. A total of four notes contained mentions of all five classes, while 288 notes mentioned none.

Additionally, Fig. 1c presents a pie chart showing how many mobility classes each section contains. As shown, 57% of sections mention no mobility classes, 23.1% mention one, 12.8% mention two, 5.8% mention three, 1.2% mention four, and 0.3% mention all five classes. Similarly, Fig. 1d illustrates how many mobility classes appear in

Characteristic	Site 1	Site 2	Site 3	Overall
Notes, <i>n</i>	200	200	200	600
Patients, <i>n</i>	92	58	107	256
Age, mean (SD)	74.3 (14.2)	74.3 (12.4)	78.0 (12.9)	75.6 (13.2)
Sex, <i>n</i> (%)				
Male	42 (45.7)	28 (48.3)	49 (45.8)	119 (46.5)
Female	50 (54.3)	30 (51.7)	58 (54.2)	137 (53.5)
Race, <i>n</i> (%)				
Black	0 (0.0%)	1 (1.7%)	0 (0.0%)	1 (0.4%)
Asian	1 (1.1%)	0 (0.0%)	0 (0.0%)	1 (0.4%)
White	89 (96.7%)	55 (94.8%)	107 (100.0%)	250 (97.7%)
Other	1 (1.1%)	0 (0.0%)	0 (0.0%)	1 (0.4%)
Refusal	1 (1.1%)	2 (3.4%)	0 (0.0%)	3 (1.2%)
Background, <i>n</i> (%)				
Hispanic	2 (2.2%)	0 (0.0%)	1 (0.9%)	3 (1.2%)
Non-Hispanic	89 (96.7%)	56 (96.6%)	105 (98.1%)	249 (97.3%)
Missing	1 (1.1%)	2 (3.4%)	1 (0.9%)	4 (1.6%)

Table 1. Characteristics of the study Population.

entire notes. It reveals that 48% of notes do not mention any mobility classes, 29.3% mention one, 9.0% mention two, 8.5% mention three, 4.5% mention four, and 0.7% reference all five.

Figure 1e–i illustrates the distribution of sections for each of the five mobility classes. For each class, we identified the three most frequently mentioned sections and grouped all remaining sections under “other”. The results showed that “history of present illness” ranks among the top three for all five mobility classes, while “assessment” appears in the top three for four of the five classes, with the exception of “Moving around using transportation”.

Baseline performance with zero-shot learning

Table 2 presents the baseline section-level performance on Task 1 (Mobility Extraction) and Task 2 (Impairment Classification). This evaluation was conducted across the three participating institutions and five mobility functional status classes. Task 1 metrics include class- and institution-specific precision, recall, and F1-scores. The micro-average F1-scores across all mobility classes are 0.663, 0.638, and 0.516 for Site 1, Site 2, and Site 3, respectively. Overall, these baseline metrics varied by class and location, with some classes performing consistently well while others presented room for improvement. They serve as a reference point against which more complex prompting strategies and configurations can be compared.

Regarding Task 2 (Impairment Classification), the class “Walking and moving” demonstrated best performance with F1-scores ranging from 0.930 to 0.964 across three sites. The class “Moving around using transportation” achieved the lowest performance with F1-scores ranging from 0.649 to 0.755 across sites. The micro-average F1-scores across all mobility classes are 0.888, 0.857, and 0.833 for Sites 1–3 respectively.

Comparison of few-shot learning and error-informed prompt refinement approaches

Figure 2a presents a comparison of section-level F1-scores for Mobility Extraction across five mobility functional status classes and three institutions under five configurations: the baseline zero-shot learning method, three few-shot learning methods (Random Selection, K-Means Error Clustering Selection, and Similarity-Based Selection), and the Error-Informed Prompt Refinement strategy. In each bar cluster, the bars represent the section-level F1-scores for a specific class and institution combination.

Overall, while most few-shot configurations underperform relative to the zero-shot baseline, the extent of this performance gap varies by both class and institution. In some cases, few-shot learning slightly outperforms the baseline, whereas in others the decline is more pronounced. One possible explanation is that longer prompts can lead the model to “forget” crucial information provided earlier in the context. In few-shot settings, the additional examples may dilute or overshadow key information like definitions of mobility classes from the beginning of the prompt.

Notably, the Error-Informed Prompt Refinement approach yields significant improvements in most scenarios, with the exceptions of the “Mobility, unspecified” class at all the three sites and the “Moving around using transportation” class at Site 3. These per-class, per-institution comparisons provide valuable insights into the conditions under which each method is most effective.

Task decomposition method performance and integration

Figure 2b illustrates the comparison of the zero-shot baseline F1-scores for Mobility Extraction against those obtained using the two task decomposition methods at section-level: (1) Single LLM with Two-Task Prompt (Chain-of-Thought Prompting) and (2) Two LLMs with Task Specialization.

As shown in Fig. 2b, neither task decomposition approach consistently outperforms the zero-shot baseline across all classes and institutions. We then integrated the two task decomposition methods with the top-

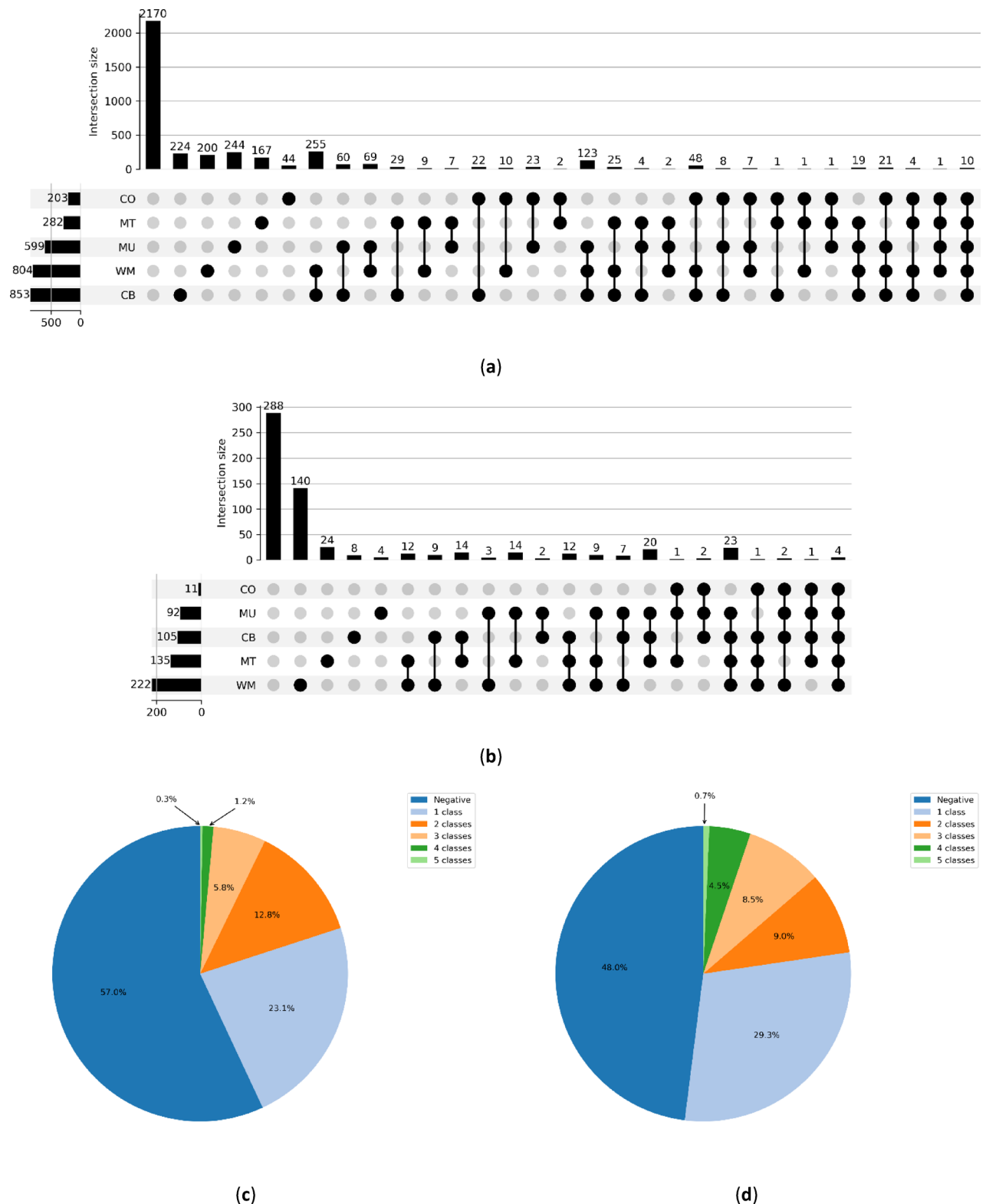


Fig. 1. Analysis of mobility classes at section- and note-levels and distributions of sections for mobility classes. The mobility classes are abbreviated as: CB (“Changing and maintaining body position”), CO (“Carrying, moving, and handling objects”), WM (“Walking and moving”), MT (“Moving around using transportation”), and MU (“Mobility, unspecified”). (a) UpSet plot of sections by mobility class combination ($n=3,810$). (b) UpSet plot of notes by mobility class combination ($n=600$). (c) Pie chart of the proportion of sections mentioning 0–5 classes: 57.0% none, 23.1% one, 12.8% two, 5.8% three, 1.2% four, 0.3% five. (d) Pie chart of the proportion of notes mentioning 0–5 classes: 48.0% none, 29.3% one, 9.0% two, 8.5% three, 4.5% four, 0.7% five. (e–i) Distribution of sections for CB, CO, WM, MT and MU, respectively, showing the three most frequent sections.

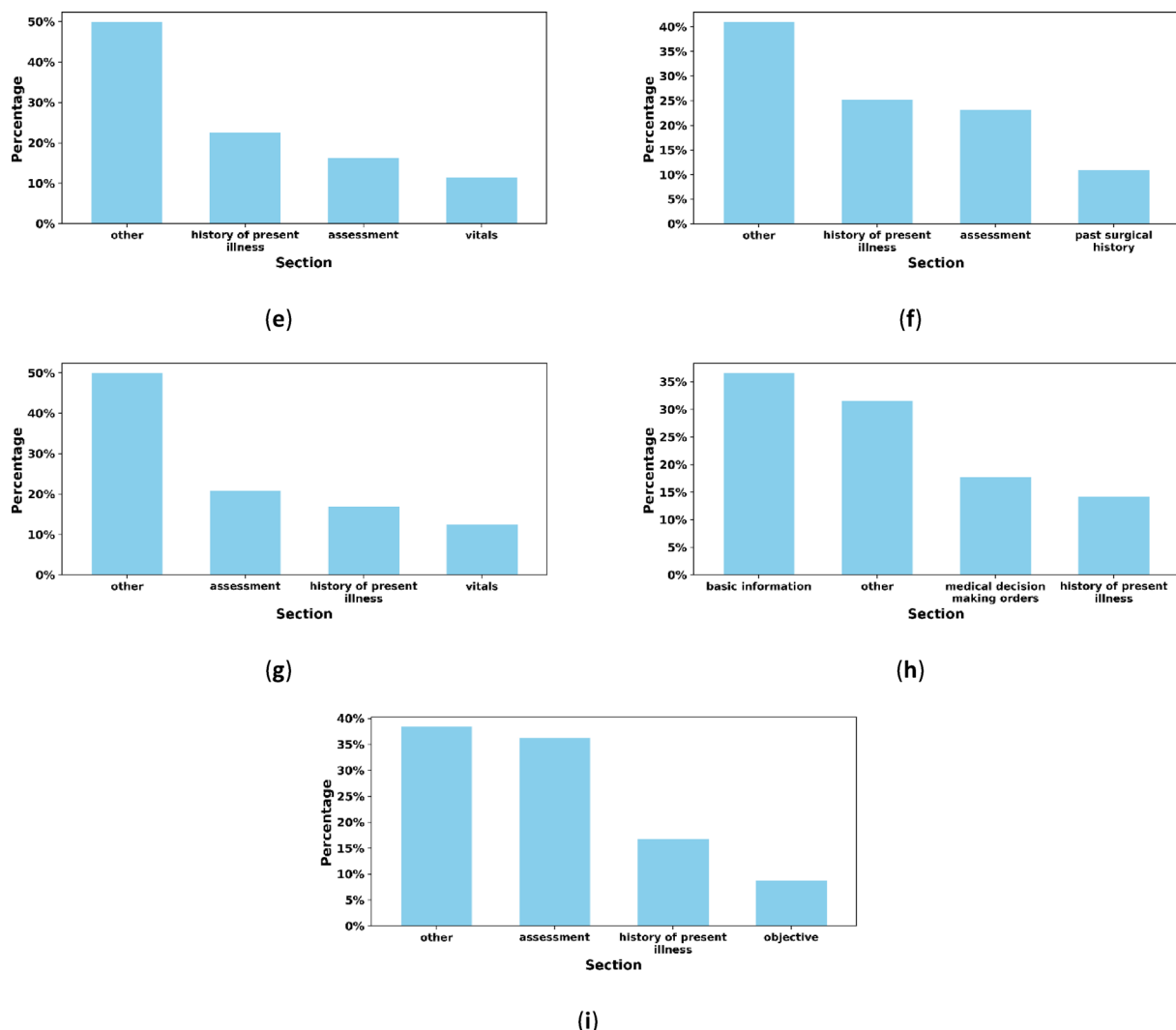


Fig. 1. (continued)

performing Error-Informed Prompt Refinement strategy. Upon re-evaluation, this combined approach did not consistently surpass the performance of the Error-Informed Prompt Refinement strategy applied alone across all classes and institutions. These findings suggest that more complex configurations do not necessarily lead to better outcomes. Overall, our best configuration remains the standalone Error-Informed Prompt Refinement strategy.

Best performance with error-informed prompt refinement

The section-level performance of the Error-Informed Prompt Refinement is summarized in Supplementary Table S1. The micro-average F1-scores across all mobility classes are 0.729 [95% CI 0.701–0.758], 0.723 [95% CI 0.695–0.745], and 0.608 [95% CI 0.577–0.642] for Sites 1, 2, and 3, respectively. As anticipated, the model's performance was lower at Site 3 for this task. This is directly attributable to the sparser and less explicit nature of its unrestricted clinical notes compared to the high-prevalence, specialized notes from Sites 1 and 2.

Regarding Task 2 (Impairment Classification) with the Error-Informed Prompt Refinement, notably, the “Walking and moving” class performed best, with F1-scores between 0.913 and 0.957 across the three sites. The micro-average F1-scores across all mobility classes are 0.832 [95% CI 0.800–0.863], 0.814 [95% CI 0.773–0.847], and 0.785 [95% CI 0.750–0.820] for sites 1, 2, and 3, respectively. Performance at Site 3 was comparable to Sites 1 and 2, with overlapping 95% confidence intervals demonstrating robust transfer for this task. This is likely because Task 2 is a classification task performed after a mobility class is identified, making it less affected by the sparse documentation at Site 3 than the initial extraction (Task 1).

Table 3 presents a comprehensive summary of the section-level, note-level and patient-level performance with the Error-Informed Prompt Refinement configuration for both tasks, aggregated across all institutions under “strict” evaluation criteria. The micro-average F1-scores for Task 1 and Task 2 were 0.695 [95% CI 0.675–0.713] and 0.815 [95% CI 0.793–0.835] at the section-level, 0.819 [95% CI 0.801–0.837] and 0.849 [95% CI 0.830–0.869] at the note-level and 0.876 [95% CI 0.858–0.894] and 0.897 [95% CI 0.878–0.917] at the patient-

Class	Site 1			Site 2			Site 3		
	Task 1		Task 2	Task 1		Task 2	Task 1		Task 2
	P	R	F1	P	R	F1	P	R	F1
Changing and maintaining body position	0.745 (0.676,0.796)	0.802 (0.747, 0.855)	0.772 (0.722, 0.812)	0.837 (0.789, 0.877)	0.789 (0.731, 0.842)	0.699 (0.639, 0.751)	0.627 (0.547, 0.702)	0.605 (0.514, 0.687)	0.665 (0.590, 0.735)
Carrying, moving, and handling objects	0.589 (0.491,0.692)	0.768 (0.655, 0.868)	0.667 (0.575, 0.751)	0.894 (0.819, 0.949)	0.602 (0.442, 0.726)	0.589 (0.468, 0.676)	0.577 (0.452, 0.694)	0.273 (0.062, 0.500)	0.545 (0.167, 0.846)
Walking and moving	0.584 (0.517, 0.646)	0.989 (0.977, 1.000)	0.734 (0.678, 0.781)	0.964 (0.946, 0.978)	0.997 (0.989, 1.000)	0.703 (0.660, 0.741)	0.542 (0.494, 0.589)	0.214 (0.182, 0.246)	1.000 (1.000, 0.394)
Moving around using transportation	0.688 (0.517, 0.833)	0.611 (0.441, 0.769)	0.647 (0.500, 0.763)	0.649 (0.417, 0.800)	0.771 (0.658, 0.883)	0.771 (0.666, 0.857)	0.771 (0.636, 0.893)	0.931 (0.882, 0.972)	0.849 (0.793, 0.898)
Mobility, unspecified	0.341 (0.290, 0.390)	0.905 (0.861, 0.946)	0.495 (0.439, 0.547)	0.872 (0.834, 0.908)	0.960 (0.928, 0.984)	0.521 (0.458, 0.576)	0.357 (0.301, 0.413)	0.252 (0.201, 0.309)	0.730 (0.651, 0.809)
Average	0.534 (0.496, 0.565)	0.874 (0.846, 0.898)	0.663 (0.630, 0.689)	0.888 (0.863, 0.909)	0.880 (0.855, 0.904)	0.638 (0.614, 0.661)	0.500 (0.473, 0.527)	0.381 (0.352, 0.411)	0.798 (0.764, 0.833)

Table 2. Performance of task 1 (Mobility extraction) and task 2 (Impairment classification) across mobility classes and institutions with zero-shot learning at section-level* . *P: Precision, R: Recall, F1: F1-score. 95% confidence intervals are shown in parentheses.

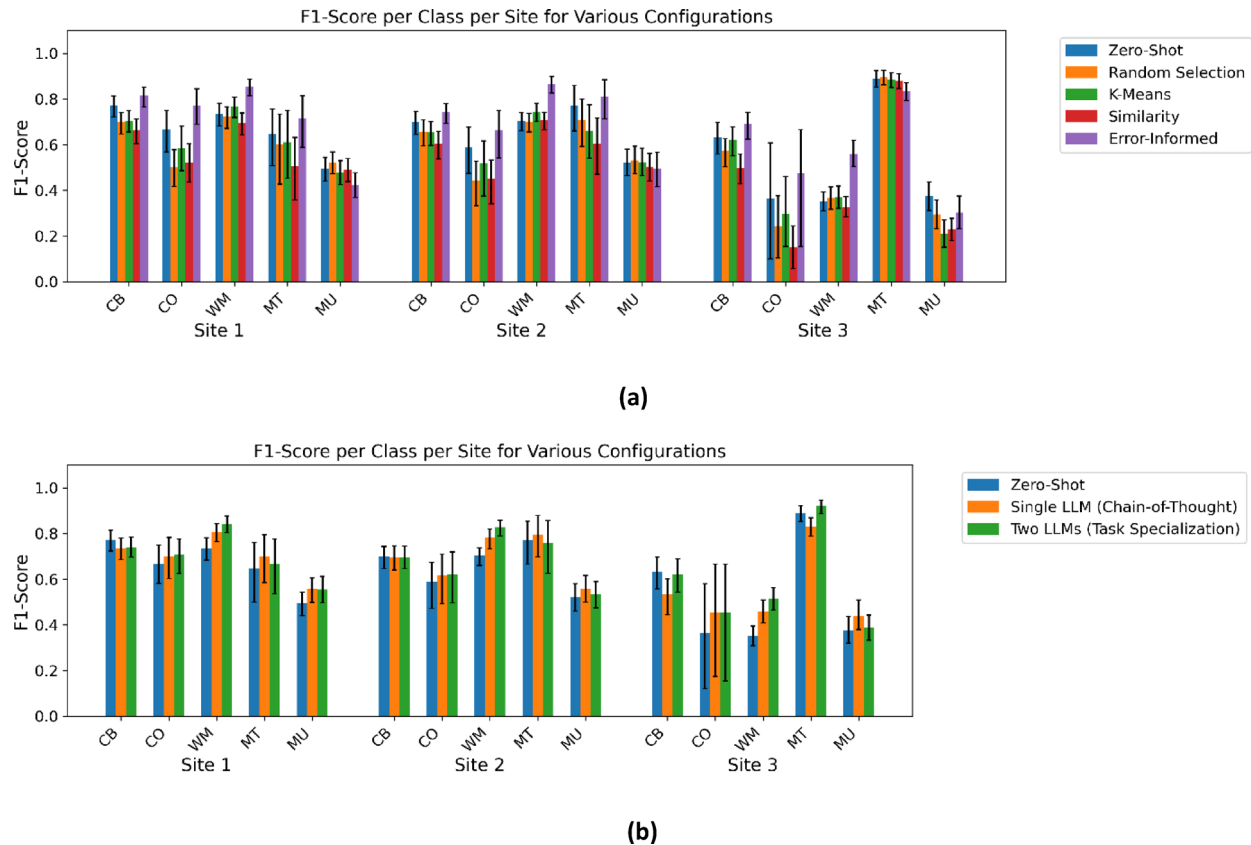


Fig. 2. Section-level F1-scores with 95% CIs for mobility extraction across five functional status classes and three institutions under different prompt and task configurations. **(a)** Comparison of zero-shot baseline, three few-shot learning strategies and error-informed prompt refinement approach. **(b)** Comparison of zero-shot baseline and two task decomposition setups. (CB: Changing and maintaining body position, CO: Carrying, moving, and handling objects, WM: Walking and moving, MT: Moving around using transportation, MU: Mobility, unspecified).

Class	Evaluation criteria	Section-level		Note-level		Patient-level	
		Task 1	Task2	Task 1	Task2	Task 1	Task 2
Changing and maintaining body position	Strict	0.757 (0.727, 0.783)	0.826 (0.802, 0.851)	0.854 (0.822, 0.883)	0.897 (0.869, 0.920)	0.901 (0.869, 0.930)	0.935 (0.907, 0.960)
	Relaxed	0.867 (0.848, 0.885)	0.866 (0.847, 0.885)	0.930 (0.908, 0.950)	0.929 (0.906, 0.948)	0.956 (0.934, 0.974)	0.951 (0.926, 0.971)
Carrying, moving, and handling objects	Strict	0.701 (0.636, 0.756)	0.817 (0.745, 0.875)	0.758 (0.682, 0.823)	0.829 (0.755, 0.892)	0.771 (0.696, 0.839)	0.831 (0.763, 0.892)
	Relaxed	0.877 (0.829, 0.918)	0.878 (0.829, 0.919)	0.909 (0.856, 0.952)	0.902 (0.848, 0.948)	0.907 (0.851, 0.949)	0.917 (0.861, 0.958)
Walking and moving	Strict	0.788 (0.754, 0.820)	0.934 (0.914, 0.950)	0.869 (0.837, 0.898)	0.931 (0.908, 0.952)	0.900 (0.871, 0.930)	0.938 (0.906, 0.964)
	Relaxed	0.948 (0.934, 0.961)	0.951 (0.937, 0.964)	0.982 (0.971, 0.991)	0.984 (0.975, 0.993)	0.989 (0.978, 0.998)	0.982 (0.968, 0.995)
Moving around using transportation	Strict	0.812 (0.778, 0.844)	0.636 (0.557, 0.711)	0.879 (0.844, 0.910)	0.670 (0.588, 0.746)	0.898 (0.858, 0.934)	0.777 (0.691, 0.850)
	Relaxed	0.877 (0.847, 0.905)	0.730 (0.655, 0.797)	0.939 (0.912, 0.963)	0.750 (0.667, 0.822)	0.955 (0.927, 0.980)	0.824 (0.746, 0.891)
Mobility, unspecified	Strict	0.428 (0.380, 0.471)	0.633 (0.576, 0.683)	0.711 (0.669, 0.749)	0.762 (0.720, 0.804)	0.852 (0.814, 0.892)	0.896 (0.858, 0.931)
	Relaxed	0.833 (0.801, 0.865)	0.810 (0.775, 0.845)	0.916 (0.892, 0.938)	0.903 (0.878, 0.926)	0.963 (0.942, 0.981)	0.966 (0.943, 0.984)
Average	Strict	0.695 (0.675, 0.713)	0.815 (0.793, 0.835)	0.819 (0.801, 0.837)	0.849 (0.830, 0.869)	0.876 (0.858, 0.894)	0.897 (0.878, 0.917)
	Relaxed	0.890 (0.878, 0.902)	0.878 (0.864, 0.893)	0.942 (0.931, 0.952)	0.925 (0.913, 0.936)	0.962 (0.952, 0.971)	0.948 (0.936, 0.960)

Table 3. F1-score of two tasks across mobility classes aggregated across all institutions with error-informed prompt refinement under strict vs. relaxed evaluation criteria*. *F1: F1-score. 95% Confidence intervals are shown in parentheses. Strict: Metrics calculated without considering reasonable inference as correct. Relaxed: Metrics calculated with considering reasonable inference as correct.

level, respectively. This aggregation effect demonstrates that while the model may be less precise on any single section, its ability to detect the correct mobility functional status in the note or for the overall patient is robust.

Error analyses

This section presents major error themes identified during our analysis. We highlight key areas for refinement and inform future enhancements to model performance and reliability. Major error themes include missing functional status context, lack of medical knowledge, ambiguous definition, issues of certainty, exclusion and inference. Inference errors were the dominant type (~75%), reflecting cases where the model inferred functional limitations without explicit evidence.

- **Inference (~75%):** Inference errors are a common source of false positives. For example, when a section mentions “right lower leg pain that began approximately 3 years ago with activities”, the LLM inferred an impairment in “Changing and maintaining body position”. Although clinically reasonable, such inferences were excluded during annotation because there was no direct mention of a limitation in that mobility class. Similarly, phrases like “right hand pain, hand swelling” led the model to infer an impairment in “Carrying, moving and handling objects”, and “has more knee pain with the exercises” prompted an inference of impairment in “Walking and moving”. In these cases, the absence of explicit evidence for a functional limitation necessitated their exclusion from impaired status assignments.

Another error pattern arises from the handling of non-specific symptoms. When sections mention term such as “dizziness”, “headaches”, “vertigo”, “weakness”, or “debility”, the LLM often assigned an impairment in the “Mobility, unspecified” class. However, in the absence of a direct reference to a functional limitation (e.g., difficulty ambulating or transferring), our annotation guidelines mandate that such general symptoms be excluded from mobility impairment classification. This overgeneralization by the LLM contributes to false positives and decreases the precision for the “Mobility, unspecified” class. Most inference errors originated in the “history of present illness” section, where symptoms frequently appear without explicit functional implications.

- **Ambiguous Definitions (~10%):** Some mobility concepts share similar semantics, leading to overlapping or inconsistent classifications. For instance, repeated falls were frequently classified by the LLM as impairments in “Walking and moving”. However, our clinical rationale dictated that such cases should be classified under “Changing and maintaining body position” to better capture issues with postural control. Similarly, mentions of wheelchair use were often split between “Walking and moving” and “Moving around using transportation”. To enhance clarity and consistency, we consolidated wheelchair-related instances exclusively under “Moving around using transportation”. These errors were concentrated in “assessment” and “physical examination” sections, where clinicians tend to document falls, gait descriptions, and assistive device use.
- **Exclusion (~10%):** Documentation pertaining to treatment plans, discussions, suggestions, instructions, recommendations, goals or advice is generally not considered indicative of a patient’s mobility functional status. This rule applies unless such documentation explicitly mentions that the patient is expected to improve in a specific mobility class, which in turn implies that an impairment exists. For example, if a note states that “the patient will improve in walking ability with physical therapy”, this suggests an underlying impairment in that area and should be annotated accordingly. Otherwise, routine mentions of treatment-related content are excluded from contributing to the functional status classification. However, we observed that the LLM sometimes struggles to exclude such content even when the prompt explicitly directs it to do so. These errors primarily occurred in “impression/report/plan” sections due to future goals or therapy instructions.
- **Limitations in Medical Knowledge (~3%):** LLMs trained on general corpora may lack the nuanced medical expertise required for certain assessments. For instance, the Focus on Therapeutic Outcomes (FOTO) system is a specialized web-based assessment where a low final score (ranging from 0 to 100) indicates impairment in the “Mobility, unspecified” class. Similarly, the Patient Specific Functional Scale (PSFS) is widely used to capture patient-specific reports of functional ability. In clinical practice, a low current PSFS score or an expectation of future improvement serves as a direct indicator of impaired functional status in the “Mobility, unspecified” class. The Timed Up and Go (TUG) Test is a standardized assessment for evaluating mobility and the risk of falls. However, our analysis revealed that LLM does not consistently integrate these quantitative assessments into its interpretation of clinical notes. As a result, numerical scores or standardized test results might be inferred indirectly.
- **Missing Definition (~2%):** The class “Mobility, unspecified” remained problematic across tasks. This challenge arises primarily from the lack of a precise ICF definition for this class, rendering its scope inherently ambiguous. In practice, this class is meant to capture general aspects of mobility like everyday activities and exercise, but it lacks clear boundaries. For example, when a clinical note states, “patient returned fully to yoga,” it implies an impairment within this class.

As noted in inference error analysis, some false positive inference cases are clinically reasonable to be considered correct (i.e., true positive). Therefore, we also evaluated the performance of the LLM after treating such reasonable inferences as correct. Table 3 also presents a comprehensive summary for both task 1 and task 2, aggregated across all institutions under this assumption (“relaxed” evaluation criteria). The performance was improved compared to the original results; the micro-average F1-score for task 1 and task 2 at the section-level increased to 0.890 and 0.878, respectively (originally 0.695 and 0.815), 0.942 and 0.925 at note-level, 0.962 and 0.948 at patient-level.

Trustworthiness analyses

LLMs used in clinical contexts must demonstrate trustworthiness across several dimensions, including reliability, safety, and robustness. In this section, we evaluate our approach and identify how our methods and configurations contribute to these attributes.

- **Reliability:** We configured the LLM with a temperature setting of 0, ensuring deterministic responses. By eliminating randomness in the generation process, the model's outputs are reproducible: repeated queries yield consistent results. Additionally, our error analysis revealed that although the LLM does make mistakes, these errors are generally “reasonable” rather than erratic or nonsensical. For instance, the model might misclassify a borderline case of mobility impairment, but it does not invent implausible conditions. Such understandable, bounded errors indicate that the model's performance is stable and predictable, which is critical for building clinical trust.
- **Safety:** Maintaining patient privacy and data confidentiality is paramount in healthcare applications. To this end, we deployed the LLM on secure local servers, preventing any transfer of protected health information (PHI) to external environments. By avoiding reliance on remote or third-party cloud services, we mitigated the risk of unintended data exposure and ensure compliance with privacy regulations. This closed-loop infrastructure provides a safer environment for processing sensitive clinical notes.
- **Generalizability:** A trustworthy LLM must demonstrate robust performance across various clinical contexts. Although the training dataset was sourced from a single site, our analysis shows that refinements—such as introducing examples or error-informed instructions—improved the model's performance not only at the original site but also at two additional independent sites. This form of transfer learning highlights the model's adaptability and generalizability, suggesting that improvements identified in one environment can successfully translate to others. Consequently, the LLM remains effective under slightly shifted data distributions, thereby increasing its overall trustworthiness as a tool for broad clinical application.

By ensuring reproducibility, logical consistency, patient data privacy, and cross-institutional adaptability, we underscore the trustworthiness of our LLM-based annotation pipeline.

Discussion

This study demonstrates the feasibility of employing LLMs with well-constructed prompt engineering to automatically extract and standardize mobility functional status information from unstructured EHR data. By leveraging an open-source LLM (Llama 3) alongside tailored prompt engineering strategies (zero-shot, few-shot, and error-informed prompt refinement), we developed a pipeline that reliably identifies and classifies mobility-related expressions across diverse clinical note sections. Our evaluations across three healthcare institutions reveal that, while performance varies among different mobility classes, the refined prompt strategy significantly enhances overall accuracy and consistency. The Error-Informed Prompt Refinement configuration was designed to address specific shortcomings observed in the baseline model by incorporating feedback from previous errors. By analyzing misclassifications and instances of low precision, the model was able to adapt its prompts to better clarify ambiguous cases. This adjustment was particularly beneficial for classes that initially suffered from high variability, such as “Carrying, moving, and handling objects” and “Walking and moving” in the Mobility Extraction task. For Impairment Classification, while the top-performing class (“Walking and moving”) maintained high performance, the refinement process contributed to a more consistent performance for classes like “Changing and maintaining body position”. The narrower range of F1-scores (0.793–0.851) in this setting suggests that error-informed adjustments helped reduce performance fluctuations across institutions.

The Error-Informed method sometimes exhibited a trade-off where enhancements in precision for some classes came at the expense of recall. For example, in Task 1, although the precision improved for the class “Mobility, unspecified”, the corresponding recall dropped, which in turn affected the F1-score. In our error analysis, we noted that the baseline model frequently inferred a patient's future goal as “mobility mentioned” (a false positive). We added a rule to the prompt to exclude such terms. While this rule successfully increased precision, it also created new false negatives (reducing recall) in cases where a clinician documented a goal as well as the current mobility functional status within the same section. It appears that when the section contains information we explicitly ask the LLM to exclude (like a treatment plan), the model sometimes ignores the entire section without checking other parts for valid information, leading to the observed drop in recall for that class.

Portability of the model across different healthcare institutions remains an important area for further exploration. Our prompt refinement was conducted using notes from Site 1, which contain a high volume of explicit mobility information. We then validated this model on both a similar dataset (Site 2) and on unrestricted clinical notes (Site 3). The results show that the model can transfer to this more difficult setting, but its performance is understandably decreased by the fact that mobility information in general notes is less frequent, less detailed, and often documented incidentally. However, this drop was not observed for Impairment Classification, where performance remained robust, demonstrating successful transfer of the classification logic. This suggests that while finding sparse mobility information (Task 1) at Site 3 is challenging, classifying it once found (Task 2) is a simpler, more transferable task.

Several practical technical insights emerge from these experiments. First, few-shot prompting did not consistently improve over the zero-shot baseline, suggesting that longer prompts and additional examples may dilute key class definitions. Second, targeted error-informed prompt refinement produced the most reliable gains, but our error analysis highlights a precision–recall trade-off when exclusion rules are tightened. Finally, more complex configurations (task decomposition and combined setups) did not consistently outperform the simpler refined prompt, indicating that incremental prompt adjustments may be more effective than adding architectural complexity for this task.

In future work, we aim to test different combinations of development and validation, and also expand data sources with more diverse patient populations, clinical note structures, and documentation styles. This would allow us to systematically assess model robustness and understand the impact of data heterogeneity on performance metrics. Additionally, while Table 1 provides essential demographic context, our analysis did not include a detailed extraction of patient comorbidities or overall health conditions. Future work should investigate the impact of comorbidity burden on both mobility status and model performance.

A limitation of this study is the lack of a direct comparison with conventional clinical NLP systems like cTAKES²⁷ or CLAMP²⁸. Such frameworks typically require extensive implementation effort, including complex pipeline development, rule-writing, and model training. In contrast, our “zero-shot” results reflect performance achieved with negligible implementation overhead. Specifically, our “Error-Informed Prompt Refinement” yielded a robust patient-level F1-score of 0.876, highlighting a highly data-efficient alternative to the heavy infrastructure often required by traditional pipelines.

Another limitation of our approach is that we did not fine-tune the LLM on the annotated dataset. Although fine-tuning has the potential to improve performance, our annotated dataset was relatively small, and fine-tuning large models typically requires substantial amounts of labeled data to avoid overfitting. In addition, fine-tuning can be computationally expensive and may increase the risk of overfitting to specific institution documentation styles. Instead, we relied on prompt-engineering strategies that can be more data-efficient and flexible. However, future research could explore lightweight fine-tuning methods (e.g., LoRA or adapter-based approaches) if larger annotated corpora become available.

Moreover, the deterministic configuration and secure local deployment of the model address key concerns regarding reproducibility and patient data privacy, further reinforcing the pipeline’s trustworthiness. Despite challenges, such as handling ambiguous language and balancing precision with recall, the findings indicate that LLMs can serve as scalable and generalizable tools for clinical applications. Future work may further explore additional domains of trustworthiness, such as fairness and interpretability, to provide an even more comprehensive assessment of the model’s reliability and suitability for clinical environments.

Longitudinal validations in real-world settings will be essential for fully integrating these methods into clinical decision-making processes, ultimately contributing to improved patient care and personalized intervention strategies. Our work provides the foundational step for multiple high-impact downstream applications. The structured output can be directly integrated as features into predictive models for outcomes such as fall-risk prediction, as well as predicting the likelihood of hospitalization or institutionalization. Furthermore, this pipeline enables the automated, longitudinal tracking of mobility status from notes. This offers a scalable method for monitoring rehabilitation progress or identifying patients who may benefit from personalized interventions.

At the clinical decision support level, this automated assessment can help optimize resource allocation by identifying patients who may require a formal physical therapy consultation, home safety evaluation, or assistive devices, even if mobility was not the primary complaint of their visit. Finally, by converting millions of unstructured notes into a structured format, this method unlocks EHR data for large-scale research on mobility, aging, and its direct impact on quality of life. This bridges the gap between unstructured clinical documentation and actionable, data-driven geriatric care.

Methods

Data preparation

This study included three institutions: an academic referral medical center and two community-based practices in the upper midwest. Legal contracts between the institutions preclude direct comparisons of data; therefore, the sites are designated by number (Sites 1–3). The annotation guidelines were developed using definitions from ICF to annotate mobility functional status data in clinical notes. We considered five mobility functional status classes, each comprising one or more ICF subclasses. Table 4 lists these classes alongside their corresponding ICF codes and subclasses.

We restricted our analysis to five primary mobility classes, rather than using their more granular ICF subclasses. This decision was driven by two practical considerations. First, our annotation process revealed a high degree of semantic overlap among the ICF subclasses, making it difficult to obtain reliable annotation for those narrower categories. As a result, we consolidated the ICF-based taxonomy into five major classes as defined in ICF to enhance labeling consistency and reduce confusion during both manual annotation and automated extraction. Second, LLMs may struggle with overly fine-grained distinctions in clinical text, especially if subclass definitions overlap or are ambiguously documented.

To facilitate focused and coherent annotation, each clinical note was divided into sections. Clinical notes at Site 1 include a variety of sections, such as History of Present Illness, Past Medical/Surgical History, Physical Examination, Diagnosis and more. We chose section-level analysis for several reasons. First, clinical notes can be lengthy, and feeding entire notes into a LLM risks exceeding context length limits, which could potentially truncate important information. Second, sections typically focus on specific aspects of a patient’s health (e.g., History of Present Illness, Assessment), which makes them natural, semantically coherent units for targeted extraction tasks. Third, section-level segmentation facilitates more precise few-shot prompts and helps the model focus on local contextual cues relevant to mobility. Finally, working at the section-level can improve interpretability, since clinicians often review notes by scanning sections, allowing direct alignment of the LLM’s output with established clinical documentation practices.

Within each section, a trained annotator manually identified and labeled mobility classes with related expressions and assigned the appropriate impairment statuses. The annotator assigned “Impaired” if the section describes difficulties, challenges, limitations, potential issues, or impairments, and “Unimpaired” if the section explicitly describes normal function or abilities. If a section did not receive either “Impaired” or “Unimpaired,”

Class	ICF code	ICF subclass
Changing and maintaining body position	d410	Changing basic body position
	d415	Maintaining body position
	d420	Transferring oneself
	d429	Changing and maintaining body position, other specified and unspecified
Carrying, moving and handling objects	d430	Lifting and carrying objects
	d435	Moving objects with lower extremities
	d440	Fine hand use
	d445	Hand and arm use
	d446	Fine foot use
	d449	Carrying, moving and handling objects, other specified and unspecified
Walking and moving	d450	Walking
	d451	Going up and down stairs
	d455	Moving around
	d460	Moving around in different locations
	d465	Moving around using equipment
	d469	Walking and moving, other specified and unspecified
Moving around using transportation	d470	Using transportation
	d475	Driving
	d480	Riding animals for transportation
	d489	Moving around using transportation, other specified and unspecified
Mobility, unspecified	d499	Mobility, unspecified

Table 4. Five mobility functional status classes and corresponding ICF codes and subclasses.

this indicated that there was no mobility-related information present in that section. Additional details on the annotation process and guidelines can be found in the Supplementary Section.

In total, 600 notes were annotated. Of these, 200 were physical therapy (PT) or occupational therapy (OT) notes from Site 1, 200 were PT/OT notes from Site 2, and 200 were unrestricted clinical notes from Site 3. The PT/OT notes (Sites 1 and 2) originate from services specializing in functional status and thus contain a high prevalence of mobility-related information, providing a focused dataset for model development and prompt refinement. In contrast, the unrestricted clinical notes from Site 3 represent a more general clinical setting. This design allows for a robust evaluation of the model’s generalizability, testing its ability to transfer from a specialized context to a broader clinical environment. After splitting the notes into sections, the dataset comprised 3,810 Sects. (1,153 in Site 1, 1,075 in Site 2, and 1,582 in Site 3). We found that 51.97% sections from Sites 1 and 2 mentioned at least one mobility class, compared to only 30.47% sections from Site 3. Each section contained a mean of 826 characters, with section lengths averaging 932 characters in Site 1, 847 in Site 2, and 734 in Site 3. A single section may contain information relevant to multiple mobility classes or none.

Experiment

For prompt development, 200 sections were randomly selected from Site 1 to form a training dataset; the remaining sections served as the test set. Performance was evaluated using precision, recall, and F1-score metrics, both class-specific and institution-specific, across various experimental configurations.

An LLM-based annotation pipeline was developed to extract mobility functional status information. The pipeline consists of three core components: (1) the LLM, (2) a task-specific prompt combined with a clinical note section, and (3) a post-processing step that converts the LLM’s text output into structured data. As shown in Supplementary Fig. S2, the process begins with dividing the clinical notes into distinct sections. Each section is then embedded within a task-specific prompt explicitly designed for mobility functional status extraction. The LLM processes the prompt and generates a textual response based on its analysis and interpretation, and finally, a post-processing step transforms the LLM’s generative text output into structured objects. Specifically, for each section, the LLM produces, for each of the five mobility classes, a single label: “Impaired” if there is evidence of limitation, “Unimpaired” if normal function is described, or “None” if the section contains no information about that class.

Llama 3, an advanced open-source LLM developed by Meta AI, was employed for this task. Llama 3 provides enhanced efficiency and performance. Integrating Llama 3 into a local server environment ensured both patient data privacy and the use of powerful computational resources for annotation tasks. This setup allowed for the handling of sensitive medical information within a secure infrastructure. The open-source nature of Llama 3 further facilitated customization and adaptability to specific project requirements.

Performance evaluation

We calculate performance at the class level. For each mobility class, we first evaluate Mobility Extraction by considering any section labeled “Impaired” or “Unimpaired” for that class as “Mentioned” and all other sections as “Not Mentioned.” Treating “Mentioned” as the positive class and “Not Mentioned” as the negative class, we

compute precision, recall, and F1-score by comparing the LLM's predicted Mentioned/Not Mentioned labels against the human annotations for that class. Micro-average and bootstrap confidence interval were calculated to guide the performance comparison. Next, for Impairment Classification, we restrict our analysis to those sections confirmed as "Mentioned." Within this subset, the LLM's output, distinguishing between "Impaired" and "Unimpaired", is compared to the human reference label. In other words, once a section is known to address the mobility class, we assess whether the LLM's choice of "Impaired" versus "Unimpaired" matches the gold-standard annotation, and we calculate an F1-score for each class with "Impaired" as the positive label. It is important to note that our evaluation possesses a hierarchical structure: sections are nested within clinical notes, which are in turn nested within patients. Our section-level metrics are calculated by treating each section as an individual observation, a standard practice in many NLP classification tasks. We also report aggregated performance at the note- and patient-levels to provide a multi-faceted view of model performance. To assess the statistical robustness of our metrics, we calculated 95% confidence intervals (CIs) for all reported metrics.

Prompting engineering

Zero-shot learning

In the zero-shot configuration, the prompt includes only the core task description, the definitions of all relevant mobility classes, a general instruction, and a final question. It does not provide any examples. Under these conditions, the model relies solely on the given prompt, without guidance from sample inputs or outputs. Evaluating zero-shot performance establishes a baseline against which subsequent methods can be compared. The detailed prompt can be found in the Supplementary.

Few-shot learning

Few-shot learning leverages a small number of example cases embedded in the prompt to improve the model's reasoning and output quality. Unlike zero-shot prompting, few-shot prompts use illustrative examples, complete with detailed explanations, to better convey the task's underlying reasoning steps to the model.

In this study, a five-shot configuration was used, with the selected examples spanning a variety of scenarios to promote more nuanced understanding. Three strategies were tested for choosing these examples from the training dataset:

- **Random Selection:** Five samples are chosen at random.
- **K-Means Error Clustering Selection:** We first identified the errors that occurred under zero-shot conditions on the training dataset. We then applied the k-means clustering algorithm to group these error cases into five clusters. From the center of each cluster, we selected one representative sample, thereby focusing on common mistakes.
- **Similarity-Based Selection:** For each test case, we identified the top five most similar training samples using a K-nearest neighbors approach. These closely related examples serve as relevant reference points that closely match the characteristics of the test input.

Through these strategies, we aimed to discover which method of choosing examples leads to the greatest performance improvement, demonstrating how few-shot learning can be optimized to achieve more accurate and reliable results.

Error-informed prompt refinement

A systematic error analysis of the zero-shot results was conducted to identify patterns in the model's mistakes, such as common misclassifications or omissions. We manually examined all false positive and false negative outputs across sections in the training set, catalogued the most common sources of error, and distilled these into a set of explicit "exclusion" and "inclusion" rules. These rules were then injected as additional instructions into the prompt so that the LLM would be guided toward avoiding the same mistakes. For example, whenever the section mentioned "patient will begin home exercise program", the model tended to treat these as evidence of impairment, even though they are plans or instructions. To address this, we added explicit instructions telling the model to ignore any text that describes a treatment plan, recommendation, or exercise regimen. The detailed instructions after the refinement can be found in the Supplementary.

Task decomposition

Our overarching goal was to extract expressions related to a patient's mobility functional status and classify their impairment status (impaired vs. non-impaired). To potentially improve performance, we divided the task into two subtasks: Mobility Extraction and Impairment Classification. This decomposition enables a clearer, more focused approach and allows for specialized handling of each step. To implement this, we explored two different setups:

- **Single LLM with Two-Task Prompt (Chain-of-Thought Prompting):** A single LLM is guided sequentially through the two subtasks with a multi-step prompt. The prompt first instructs the model to assess whether the section contains mobility-related descriptions for each class, then directs it to classify the impairment status. By leveraging a chain-of-thought process, this setup encourages methodical, stepwise reasoning.
- **Two LLMs with Task Specialization:** Alternatively, this approach employs two dedicated LLMs, each focusing on a single subtask. The first LLM determines whether relevant mobility descriptions are present. The second LLM then uses those determinations to classify impairment status. By dividing responsibilities in this manner, each model can specialize its role.

Data availability

The data used for this work was from electronic health records which include identifiable data and thus cannot be made publicly available due to privacy and legal reasons. The source code can be made available by the corresponding author on reasonable request.

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Author contributions

X.L. developed the model, analyzed, and interpreted the data, drafted the initial manuscript, and revised the manuscript. H.J. performed data curation and corpus annotation. S.P. and J.S. helped with cohort identification and corpus annotation. S.P., J.S., H.J. and M.G. participated in the interpretation of the data and contributed to manuscript editing and revisions. S.S. conceptualized and designed the study, supervised data collection, model development, and analysis, interpreted the data, and reviewed and finalized the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

The research was performed in accordance with relevant guidelines and regulations. The study was approved by the Mayo Clinic Institutional Review Board and the Olmsted Medical Center Institutional Review Boards. Informed consent was obtained from all subjects included in this study.

Additional information

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Correspondence and requests for materials should be addressed to X.L.

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